

Name:	
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Honor pledge <i>"On my honor, I have neither given nor received unauthorized aid."</i> Student Signature: _____
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Subject:	Computing
Exam:	Year 9 Term1 Assessment 1 (Unit 1)
Exam Duration:	45 minutes
Teacher:	Mr Ellis

Instructions to Students:
Answer all questions by circling the correct answer on the bubble sheet provided. This exam contains 100 multiple-choice questions covering the entire Unit 1 Data Representation specification. Each question has four options: A, B, C, or D. Choose the best answer. All questions are equally weighted. There is paper at the end for working out calculations. Calculators are not permitted.

Grade 7 & 8 Secondary School Grade Conversion Matrix	
ISR Percentage Grade	Descriptive
97-100	Outstanding
90-96	Excellent
85-89	Very Good
80-84	Good
70-79	Satisfactory
60-69	Needs Improvement
0-59	Unsatisfactory

Marks:
ISR % Grade:

Name

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Unit 1 Data Representation · 100-Question Exam

Binary · Hexadecimal · Shifts · Character Sets · Sound · Images · Compression

Section 1: Binary Systems (Q1–15)

1. **Why do computers use binary?**
 - A Because binary is easier for humans to read
 - B Because computers are made of transistors that have two states: on and off
 - C Because binary allows for more colours than denary
 - D Because binary uses ten digits like the denary system
2. **How many bits are in one byte?**

A 4	B 8
C 16	D 1024
3. **What is the denary value of the binary number 10110101?**

A 169	B 181
C 197	D 203
4. **Convert the denary number 42 to binary.**

A 00101010	B 00101100
C 01010100	D 00100110
5. **What is the smallest unit of data in a computer?**

A A byte	B A bit
C A nibble	D A kilobyte
6. **What is the largest denary number that can be represented with 8 bits?**

A 128	B 255
C 256	D 1024
7. **Which is the binary representation of the denary number 100?**

A 01100100	B 01101000
C 01010100	D 01100010
8. **How many different values can be represented with 4 bits?**

A 4	B 8
C 16	D 32
9. **What is the denary value of the binary number 11111111?**

A 127	B 128
C 255	D 256
10. **Which term describes half a byte (4 bits)?**

A A bit	B A nibble
C A word	D A kilobyte
11. **What is the denary value of the binary number 00111001?**

A 45	B 57
C 63	D 71
12. **How many bytes are in a kibibyte (KiB)?**

A 1000	B 1024
C 2048	D 512

13. The number of bytes in a mebibyte can be written as 2^x . What is x ?

- | | |
|------|------|
| A 10 | B 15 |
| C 20 | D 24 |

14. Which is the binary representation of the denary number 200?

- | | |
|------------|------------|
| A 11001000 | B 11010000 |
| C 10101000 | D 11100000 |

15. What is the denary value of the binary number 01011010?

- | | |
|------|------|
| A 86 | B 90 |
| C 94 | D 98 |

Section 2: Hexadecimal (Q16–30)

16. What is hexadecimal base?

- | | |
|-----------|-----------|
| A Base 2 | B Base 8 |
| C Base 10 | D Base 16 |

17. In hexadecimal, what denary value does the letter F represent?

- | | |
|------|------|
| A 10 | B 14 |
| C 15 | D 16 |

18. Convert the hexadecimal number 2A to denary.

- | | |
|------|------|
| A 32 | B 40 |
| C 42 | D 44 |

19. Convert the denary number 27 to hexadecimal.

- | | |
|------|------|
| A 1A | B 1B |
| C 1C | D 27 |

20. Convert the binary number 11100101 to hexadecimal.

- | | |
|------|------|
| A E5 | B D5 |
| C E3 | D F5 |

21. Why is hexadecimal often used in computing instead of binary?

- A It is easier to read and less error-prone
- B It allows more colours than binary
- C It uses fewer bits per value than binary
- D It is faster for the CPU to process

22. What is the largest denary number that can be represented with two hex digits?

- | | |
|-------|--------|
| A 15 | B 255 |
| C 256 | D 1024 |

23. Convert the hexadecimal number 3C to binary.

- | | |
|-------------|-------------|
| A 0011 1100 | B 0011 1010 |
| C 0100 1100 | D 0011 1110 |

24. Convert the denary number 42 to hexadecimal.

- | | |
|------|------|
| A 2A | B 2B |
| C 2C | D 3A |

25. What is the denary value of the hexadecimal number 7F?

- | | |
|-------|-------|
| A 112 | B 127 |
| C 128 | D 255 |

26. Which is the hexadecimal equivalent of the binary number 10101111?

- | | |
|------|------|
| A AF | B BF |
| C 9F | D AE |

27. In hexadecimal, what comes after 19?

- | | |
|-------------|-------------|
| A 20 | B 1A |
| C 1B | D 21 |

28. What is the denary value of the hexadecimal number FF?

- | | |
|--------------|--------------|
| A 15 | B 128 |
| C 255 | D 256 |

29. Which is the binary equivalent of the hexadecimal number A5?

- | | |
|--------------------|--------------------|
| A 1010 0101 | B 1010 0110 |
| C 1011 0101 | D 1001 0101 |

30. Convert the denary number 200 to hexadecimal.

- | | |
|-------------|-------------|
| A C8 | B B8 |
| C D8 | D CA |

Section 3: Binary Shifts & Two's Complement (Q31–45)

31. What is the effect of a logical left shift by one place on a binary number?

- | | |
|-----------------------|--------------------------|
| A Divides by 2 | B Multiplies by 2 |
| C Adds 2 | D Subtracts 2 |

32. What is the effect of a logical right shift by one place on a binary number?

- | | |
|--------------------------|-----------------------|
| A Multiplies by 2 | B Divides by 2 |
| C Adds 2 | D Subtracts 2 |

33. Perform a left shift by one place on the binary number 00111100.

- | | |
|-------------------|-------------------|
| A 01111000 | B 00011110 |
| C 01111100 | D 00111100 |

34. Perform a right shift by one place on the binary number 01011010.

- | | |
|-------------------|-------------------|
| A 00101101 | B 10110100 |
| C 00101010 | D 01011010 |

35. What is 45 in binary, then right shifted by one place?

- | | |
|-------------|-------------|
| A 22 | B 23 |
| C 45 | D 90 |

36. What is the effect of a right shift on an odd number?

- A** It is rounded up
- B** It is rounded down (loss of precision)
- C** It stays the same
- D** It becomes negative

37. What is two's complement used for?

- | | |
|--------------------------------------|---|
| A Converting denary to binary | B Representing negative integers in binary |
| C Adding binary numbers | D Compressing binary data |

38. What is -15 in 8-bit two's complement?

- | | |
|-------------------|-------------------|
| A 11110000 | B 11110001 |
| C 10001111 | D 00001111 |

39. What is the denary value of the two's complement number 11111011?

- | | |
|-------------|-------------|
| A -5 | B -4 |
| C -3 | D -1 |

40. What is the range of values that can be represented with 8-bit two's complement?

- | | |
|----------------------|----------------------|
| A 0 to 255 | B -127 to 127 |
| C -128 to 127 | D -128 to 128 |

41. Perform a left shift by two places on the binary number 00001111.
- | | |
|------------|------------|
| A 00111100 | B 00011110 |
| C 11110000 | D 00001111 |
42. What is the denary value of 00001111 after a left shift by two places?
- | | |
|------|-------|
| A 15 | B 30 |
| C 60 | D 120 |
43. What happens to the bits shifted off the end of a binary register?
- A They are stored in a separate register
 - B They are lost
 - C They wrap around to the other end
 - D They are added to the result
44. What is -1 in 8-bit two's complement?
- | | |
|------------|------------|
| A 10000000 | B 11111111 |
| C 00000001 | D 11111110 |
45. What is the denary value of the two's complement number 10000000?
- | | |
|--------|--------|
| A -128 | B -127 |
| C 0 | D 128 |

Section 4: Character Sets (Q46–60)

46. What is a character set?
- A A set of binary numbers used for arithmetic
 - B A defined list of characters recognised by a computer, each with a unique binary code
 - C A group of fonts used for displaying text
 - D A compression method for text files
47. How many bits does standard ASCII use per character?
- | | |
|-----|------|
| A 4 | B 7 |
| C 8 | D 16 |
48. How many characters can standard 7-bit ASCII represent?
- | | |
|-------|-------|
| A 64 | B 128 |
| C 256 | D 512 |
49. What is one limitation of the ASCII character set?
- A It cannot represent numbers
 - B It cannot represent characters from other alphabets such as Arabic or Chinese
 - C It uses too many bits per character
 - D It is not compatible with modern computers
50. What character set was designed to overcome the limitations of ASCII?
- | | |
|------------------|-----------|
| A Extended ASCII | B Unicode |
| C EBCDIC | D BCD |
51. How many bits does Unicode typically use?
- | | |
|----------------|------|
| A 7 | B 8 |
| C 8, 16, or 32 | D 64 |
52. How many characters can 16-bit Unicode represent?
- | | |
|----------|---------|
| A 128 | B 256 |
| C 65,536 | D 1,024 |
53. Which is a limitation of the ASCII character set?
- A It can only represent the English alphabet and a few symbols
 - B It cannot represent numbers
 - C It uses 16 bits per character
 - D It cannot be used on modern computers

54. If 'A' is represented by 65 in denary, what is 'C'?

- A** 66 **B** 67
C 68 **D** 69

55. How many bytes are needed to store the message "Hello" in 8-bit extended ASCII?

- A** 4 **B** 5
C 6 **D** 8

56. Extended ASCII uses 8 bits. How many characters can it represent?

- A** 128 **B** 256
C 512 **D** 1,024

57. Why can't you perform arithmetic directly on ASCII values of digits?

- A** Because ASCII only stores letters
B Because the ASCII code for '7' is different from the numeric binary value 7
C Because ASCII values are too large
D Because computers cannot add ASCII values

58. Which statement about ASCII is correct?

- A** Every character (including space and punctuation) has its own unique binary code
- B** Only letters have binary codes; spaces and punctuation are ignored
- C** All characters use the same binary code
- D** ASCII codes can only be used for numbers, not text

59. Which character set includes emojis?

- A** ASCII **B** Extended ASCII
C Unicode **D** BCD

60. How many bits are needed to represent 256 different characters?

- A** 7 **B** 8
C 16 **D** 32

Section 5: Representing Sound (Q61–75)

61. What is the name of the process where an analogue sound wave is measured at regular intervals to convert it into digital data?

- A** Modulation **B** Sampling
C Amplification **D** Compression

62. What is the sample rate?

- A** The number of bits used to represent each sample
B The number of samples taken per second
C The total duration of the recording
D The frequency of the sound wave

63. Which best describes sample resolution?

- A** The number of samples taken per second
B The number of bits used to store the value of each sample
C The length of time the recording lasts
D The volume (amplitude) of the sound wave

64. What effect does increasing the sample rate have on the sound file?

- A** It decreases the file size and improves quality
B It increases the file size and improves quality
C It decreases the file size and reduces quality
D It has no effect on file size or quality

65. What effect does increasing the sample resolution have on the sound file?

- A It decreases the file size and improves quality
- B It increases the file size and improves accuracy of each sample
- C It decreases the file size and reduces quality
- D It has no effect on file size or quality

66. Which formula correctly calculates the file size of an uncompressed sound file?

- A File size = sample rate $\times$ duration $\times$ sample resolution
- B File size = sample rate $\times$ sample resolution $\div$ duration
- C File size = duration $\times$ sample resolution $\div$ sample rate
- D File size = sample rate + duration + sample resolution

67. A sound is recorded for 2 seconds at a sample rate of 1,000 Hz with a resolution of 8 bits. What is the file size in bits?

- | | |
|---------------|---------------|
| A 2,000 bits | B 8,000 bits |
| C 16,000 bits | D 32,000 bits |

68. Same recording (2 s, 1,000 Hz, 8 bits). What is the file size in bytes?

- | | |
|---------------|---------------|
| A 500 bytes | B 1,000 bytes |
| C 2,000 bytes | D 4,000 bytes |

69. What is the typical hearing range for humans?

- | | |
|------------------------|---------------------|
| A 20 Hz to 20,000 Hz | B 0 Hz to 10,000 Hz |
| C 100 Hz to 100,000 Hz | D 1 Hz to 1,000 Hz |

70. Which device converts analogue sound signals into digital data?

- | | |
|---------------------------------------|---------------------------------------|
| A DAC (Digital-to-Analogue Converter) | B ADC (Analogue-to-Digital Converter) |
| C Amplifier | D Speaker |

71. Which device converts digital data back into an analogue sound signal?

- | | |
|---------------------------------------|---------------------------------------|
| A ADC (Analogue-to-Digital Converter) | B DAC (Digital-to-Analogue Converter) |
| C Microphone | D Amplifier |

72. What happens to the quality of a sound recording if the sample rate is too low?

- A The sound is more accurate
- B The sound is less accurate and may miss high frequencies
- C The sound file is larger
- D The sound is unchanged

73. Which of the following is a lossy audio file format?

- | | |
|--------|--------|
| A FLAC | B WAV |
| C MP3 | D ALAC |

74. What is the main advantage of lossless audio compression?

- A It produces much smaller files than lossy compression
- B It preserves the original audio exactly
- C It is faster to stream
- D It requires less storage space than lossy

75. If a sound file has a sample rate of 500 Hz, how many samples are taken in 4 seconds?

- | | |
|---------|---------|
| A 500 | B 1,000 |
| C 2,000 | D 4,000 |

Section 6: Representing Images (Q76–88)

76. What is a pixel?

- A A type of image file format
- B The smallest unit of a digital image
- C A colour code used in HTML
- D A compression algorithm for images

77. What is the resolution of an image?

- A The number of colours in the image
- B The number of pixels in the image (width × height)
- C The file size of the image
- D The number of bits per pixel

78. What is colour depth?

- A The number of pixels in the image
- B The number of bits used to represent the colour of a single pixel
- C The physical size of the image on screen
- D The file format of the image

79. How many colours can be represented with a 1-bit colour depth?

- A 1
- B 2
- C 4
- D 8

80. How many colours can be represented with a 24-bit colour depth?

- A 256
- B 65,536
- C 16.7 million
- D 1 billion

81. What is the formula for calculating the file size of an uncompressed image?

- A Width × Height × Colour Depth
- B Width + Height + Colour Depth
- C Width × Height ÷ Colour Depth
- D Colour Depth ÷ (Width × Height)

82. An image is 10 × 10 pixels with 8-bit colour depth. What is its file size in bits?

- A 100 bits
- B 800 bits
- C 1,000 bits
- D 8,000 bits

83. Same image (10 × 10, 8-bit). What is its file size in bytes?

- A 10 bytes
- B 100 bytes
- C 800 bytes
- D 1,000 bytes

84. What effect does increasing the colour depth have on the image file size?

- A It decreases the file size
- B It increases the file size
- C It has no effect on file size
- D It halves the file size

85. What effect does reducing the resolution have on an image?

- A The image becomes sharper
- B The image loses fine detail and may appear blurred
- C The image gains more colours
- D The image file size increases

86. What is metadata in an image file?

- A The pixel data itself
- B Data about the data — such as width, height, colour depth, and format
- C The compression algorithm used
- D The colour palette of the image

87. Which of the following is a lossless image format?

- A JPEG
- B PNG
- C MP3
- D MP4

88. What does RLE stand for?

- A Run Length Encoding
- B Random Loss Encoding
- C Reduced Loss Encoding
- D Real Loss Encoding

Section 7: Compression (Q89–100)

89. What is compression?

- A The process of increasing file size for better quality
- B The process of reducing file size using algorithms
- C The process of converting files to binary
- D The process of encrypting files

90. What is decompression?

- A The process of compressing a file further
- B The process of restoring compressed data to a usable format
- C The process of deleting compressed files
- D The process of copying compressed files

91. What is the main difference between lossy and lossless compression?

- A Lossy reduces file size; lossless does not
- B Lossy permanently removes some data; lossless does not
- C Lossless is faster than lossy
- D Lossy is used only for text files

92. Which of the following file formats uses lossy compression?

- | | |
|-------|--------|
| A PNG | B JPEG |
| C ZIP | D FLAC |

93. Which of the following file formats uses lossless compression?

- | | |
|-------|--------|
| A MP3 | B MP4 |
| C PNG | D JPEG |

94. How does Run Length Encoding (RLE) compress data?

- A By removing every other bit
- B By recording frequency/data pairs for runs of identical values
- C By converting binary to hexadecimal
- D By randomly discarding data

95. Using RLE, how would the row 11110000 be encoded (1 = white, 0 = black)?

- | | |
|-----------|-----------|
| A 4 1 4 0 | B 1 4 0 4 |
| C 8 1 0 | D 4 0 4 1 |

96. What is a digital artefact in the context of lossy compression?

- A A type of file format
- B A small visual mistake introduced by lossy compression
- C A compression algorithm
- D A type of metadata

97. Why is lossy compression unsuitable for compressing a computer program?

- A It would make the program run faster
- B A single missing character could stop the program from running
- C It would increase the file size
- D It would change the programming language

98. What is the effect of reducing the sample rate of an audio file?

- A It increases the file size and quality
- B It decreases the file size and permanently loses some sound data
- C It has no effect on the file
- D It increases the number of bits per sample

99. Which of the following is a benefit of compression?

- A** Increased bandwidth required
- B** Longer transmission time
- C** Less storage space required
- D** More data usage on mobile contracts

100. Which of the following is true about lossless compression?

- A** It permanently removes some data
- B** It restores the original data exactly
- C** It is used only for video files
- D** It always produces smaller files than lossy compression

End of exam · Total 100 marks · Unit 1 Data Representation

Extra space for working

